

Matthew (Man Fai) Wong

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Summary

Agentic AI and machine learning engineer with a Ph.D. in Computer Science, specializing in MCP-based agent platforms and multimodal AI systems. Experienced building LLM-powered workflows, knowledge distillation pipelines, and compact on-device computer-vision and deep-learning models, deployed from cloud agent services to edge hardware.

Experience

Machine Learning Engineer | FluxInk Technology Inc. | Canada Jul 2025 - Sep 2026

- Architected and shipped a multi-tenant MCP platform delivering handwriting recognition, OCR, and style transfer as composable tools behind a central gateway, with MCP Apps that layer interactive UIs over those tools across ChatGPT widgets, Claude connectors, and Microsoft 365 Copilot agents
- Developed a compact CNN-GRU stylus trajectory prediction model for OEM tablets that forecasts upcoming pen positions so ink renders ahead of the physical stroke, cutting the visible lag between pen tip and ink; trained with risk-aware custom loss functions to suppress prediction errors on high-curvature strokes
- Designed and prototyped agentic AI workflows for FluxInk's online AI agent services, including mode-specific tool orchestration, forced-tool routing, plan-validate-repair loops, and streaming/non-streaming execution paths to make long-running canvas, slide, and document-generation tasks more reliable
- Built a knowledge distillation and model compression pipeline (large-scale synthetic data generation and verification, distributed fine-tuning, vocabulary pruning, ONNX INT4 quantization) that shrank a multi-billion-parameter model by over an order of magnitude while preserving generation quality, enabling fully on-device generation of structured notes in multiple formats (e.g., Cornell, meeting minutes)
- Built an on-device pipeline combining OCR with custom CNN detectors for diagram structure (nodes, connectors, arrows) to convert hand-drawn flowcharts into structured Mermaid/DOT/TikZ output via deterministic graph-reconstruction logic, trained on synthetic data with ONNX-verified numerical parity

Technical Analyst, Data Science & Engineering | OOCL | Hong Kong Nov 2023 – Mar 2025

- Built a news-intelligence pipeline that scores unstructured multi-source shipping news for pricing and freight relevance, combining LLM-based entity extraction and classification with a trained LightGBM importance model, and automated daily delivery via Power Automate with Tableau dashboards and Microsoft Teams alerts
- Developed a vessel trim optimization model using LightGBM on time-series telemetry with engineering features from mechanical and operational signals (MAE below 5%), and built a React/Next.js IoT platform that ingests navigation, fuel, and sensor data to deliver fleet performance analysis and integrate the optimization model
- Containerized ML training and inference workloads with Docker, automated model promotion through Azure DevOps CI/CD pipelines, and integrated experiment tracking with MLflow and ClearML to standardize reproducible deployments across development, staging, and production environments

Education

City University of Hong Kong, Ph.D. in Computer Science Nov 2019 – Jun 2024

- Thesis: "LLM and Crowdsourced Computation for AI-assisted Programming and Its Applications"

City University of Hong Kong, B.Sc. in Computer Science Aug 2015 – Oct 2019

- Graduated with First Class Honours (Data Science Stream)

Skills

Programming: Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

Databases: StorageGrid, Azure Blob Storage, MySQL, PostgreSQL, Oracle, Trino, Chroma, FAISS, pgvector.rs

Frameworks & Tools: React, Next.js, Node.js, FastAPI, PyTorch, Transformers, CVX, MOSEK, Gurobi, Tableau, Power Automate, Docker, Kubernetes, GitLab CI/CD, Jenkins, RabbitMQ, Redis, AWS S3

MLOps & ML Infra: Kedro, ClearML, MLflow, vLLM, Ollama, Unsloth, OpenVINO, Windows ML SDK

RAG & AI Agents: LangChain, LangGraph, MCP, OpenAI Agents SDK, Dify, n8n

Languages: English, Mandarin, Cantonese

Internships

AI Lab Intern Tencent China	Aug 2018 – Jan 2019
- Conducted research on social network analysis, focusing on chat group recommendation using graph-based data - Optimized algorithms on graphs (e.g., random walk and Massey's method) to support experimental setups	
Student Trainee Octopus Holdings Limited Hong Kong	Jul 2017 – Jul 2018
- Developed backend services for Dubai RTA Nol Card clearing and settlement system using Java Spring Framework - Performed QA testing (unit tests, UAT) and managed deployments using Apache Maven and Jenkins	

Publications

Aligning Crowd-sourced Human Feedback for Reinforcement Learning on Code Generation by LLMs Man Fai Wong, Chee Wei Tan	IEEE Trans BigData'24
Automatic Feedback Generation on K-12 Students' DS Education by Prompting Cloud-based LLMs Sze Ching Evelyn Fung, Man Fai Wong, Chee Wei Tan	ACM L@S'24
Aligning Crowd-sourced Human Feedback for Code Generation with Bayesian Inference Man Fai Wong, Chee Wei Tan (Honorable Mention in Foundation Models & GenAI)	IEEE CAI'24
Chain-of-Thoughts Prompting with Language Models for Accurate Math Problem-Solving Sze Ching Evelyn Fung, Man Fai Wong, Chee Wei Tan	IEEE MIT URTC'23
Copilot for Xcode: Exploring AI-Assisted Programming for Software Development Chee Wei Tan, Shangxin Guo, Man Fai Wong, Ching Nam Hang	LLM@IJCAI'23
Natural Language Generation and Understanding of Big Code for AI-Assisted Programming: A Review Man Fai Wong, Shangxin Guo, Ching Nam Hang, Siu Wai Ho, Chee Wei Tan	Entropy'23
EuclidNet: Deep Visual Reasoning for Constructible Problems in Geometry Man Fai Wong, Xintong Qi, Chee Wei Tan	AAIML'23
The PageRank Bipartite Graph Ranking: Online Chat Group Recommendation Man Fai Wong, Ching Nam Hang, Chee Wei Tan	Online Social Networks'20
Mathematics Gamification in Mobile App Software for Personalized Learning at Scale Chee Wei Tan, Lin Ling, Pei Duo Yu, Ching Nam Hang, Man Fai Wong	IEEE ISEC'19

Awards

Honorable Mention Award in Foundation Models and Generative AI, IEEE CAI	2024
Outstanding Academic Performance Award, City University of Hong Kong	2023
Talent Development Scholarship, Hong Kong SAR Government	2019
Shanghai Commercial Bank Scholarships, Shanghai Commercial Bank	2019

Selected Awards & Projects

Hong Kong ZooHackathon - Second Prize, U.S. Consulate General HK & Macau	2019
Built an iOS computer vision prototype for whale shark identification, sighting analysis, and movement prediction using TensorFlow and open ocean data	
Microsoft Imagine Cup National Final - Runner-up, Microsoft	2018
Developed a camera-based AI assistant for veterinary diagnosis, combining image recognition, symptom reasoning, and knowledge-based recommendations	
Intel Cup Embedded System Design Invitational Contest - Second Prize, Intel	2018
Designed an embedded AI diagnostic system on Intel UP Squared Board for real-time visible-symptom analysis	
China Mobile Application Innovation Contest - Second Prize, Apple	2018
Built a Swift iOS AR makeup assistant that used facial-feature analysis to provide real-time tutorials, personalized product recommendations, and saved look previews	
hackUST - Champion of Cybersecurity, HKUST	2018
Built a multimodal mobile authentication system using gesture signals, motion data, and facial recognition	